

Monte Carlo Pricing of an Interest-Rate-Linked Structured Note

Financial Data Analytics
Lamprou Nikolaos

Contents

1. Introduction and investment context
2. Product design and modelling assumptions
 - 2.1 Structured note specification
 - 2.2 Historical interest-rate environment
 - 2.3 Interest-rate modelling framework
3. Monte Carlo simulation and baseline valuation
 - 3.1 Simulation methodology
 - 3.2 Baseline valuation results
 - 3.3 Baseline investment assessment
4. Alternative discounting scenario
 - 4.1 Methodological adjustments
 - 4.2 Valuation impact of lower discount curve
 - 4.3 Role of the discount rate in fixed-income pricing
5. Shorter maturity scenario
 - 5.1 Simulation setup for 4-year note
 - 5.2 Valuation and risk for the 4-year note
 - 5.3 Comparative assessment across scenarios
 - 5.4 Risk decomposition and portfolio perspective
6. Conclusion: implications for the investment decision

1. Introduction and investment context

The structured note under analysis pays annual coupon payments which depend on two benchmark interest rates that researchers identify as BIR1 and BIR2 while all cash flows receive discounting through a different discounting interest rate which researchers refer to as DIR. The analysis is conducted from the perspective of a junior quantitative analyst at a hedge fund specialising in fixed-income and structured

products, with the objective of assessing whether the note offers an attractive risk–return profile at issuance relative to its face value.

The primary assignment involves creating a monthly interest rate model which will generate coupon payments based on a capped-and-floored coupon scheme and then apply Monte Carlo simulation to determine the note's value through multiple testing conditions. The report begins by determining the 8 year note's fair value using a standard discounting curve (DIR) which it later revalues through an alternative discount curve (DIR_alt) while examining how halving the maturity period to 4 years impacts both valuation and risk assessment. The research combines academic modeling and simulation methods with professional hedge fund performance evaluation to analyze investment effects and interest rate exposure.

2. Product design and modelling assumptions

2.1 Structured note specification

The structured note is assumed to be issued at par with a face value normalised to 100 and to pay annual coupons over an 8-year maturity in the baseline specification. Coupon payments are linked to two benchmark rates, BIR1 and BIR2, which can be interpreted as points on the yield curve relevant for a curve-steepener style payoff, while discounting is based on the discounting interest rate, DIR. At each coupon date t , the coupon rate c_t is given by the following rule:

$$c_t = \min \{C, \max [F, \alpha(BIR1_t - BIR2_t) + \beta BIR1_t]\}.$$

Here F denotes the coupon floor, C the cap, and α and β are coefficients chosen to determine the sensitivity of the coupon to the term-structure spread and to the level of BIR1, respectively. In this analysis, the parameters are set to $\alpha = 4$, $\beta = 0.5$, a floor $F = 0.005$ (0.5 per cent) and a cap $C = 0.08$ (8 per cent), with a face value $FV = 100$. These choices generate a coupon profile that benefits from a steep yield curve (high BIR1 – BIR2) and from higher levels of BIR1, while the floor and cap control downside and upside exposure in line with common constant-maturity-swap (CMS) steepener structures used in practice.

The coupon payment at each date is simply $c_t \times FV$, and at maturity the investor receives the final coupon plus repayment of the face value. Under the baseline 8-year design, there are eight annual coupon dates. The note therefore combines exposure to the level and slope of the interest-rate term structure with principal protection at maturity, making it suitable for investors with a moderately bullish view on curve steepness and rate levels but who still value bounded coupon risk.

2.2 Historical interest-rate environment

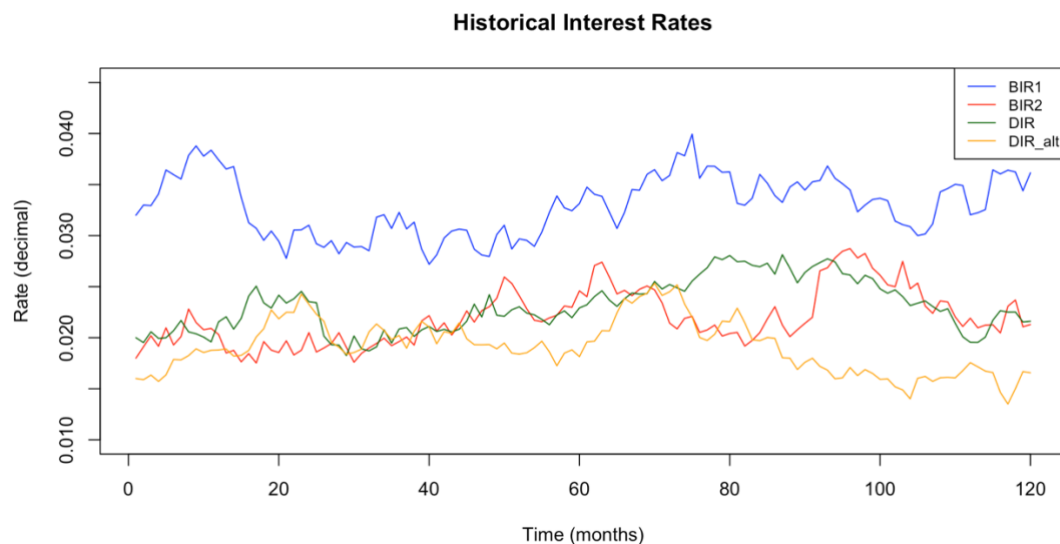
The analysis uses the Data2.csv file supplied with the assignment, which contains historical monthly observations for BIR1, BIR2, DIR, and an alternative discounting curve DIR_alt. Descriptive statistics for these series are reported in Table 1.

Table 1: Descriptive statistics for historical monthly interest rates (in decimal form).

Series	Mean	Std. dev.	Min	Max
BIR1	0.03298	0.00296	0.02719	0.03994
BIR2	0.02197	0.00271	0.01752	0.02873
DIR	0.02306	0.00254	0.01825	0.02813
DIR_alt	0.01911	0.00261	0.01350	0.02533

BIR1 has the highest average level, with a mean of approximately 3.30 per cent, while BIR2 and DIR trade around 2.20–2.30 per cent; DIR_alt has a lower mean of around 1.91 per cent. All four series exhibit relatively tight ranges and modest standard deviations, with no extreme outliers, which supports the use of mean-reverting short-rate models calibrated to this environment.

Figure 1: Historical interest rates (BIR1, BIR2, DIR, DIR_alt) over the sample period.



From a portfolio-management perspective, the persistent positive spread between BIR1 and BIR2 implies a structurally steep curve over the sample window, which is economically consistent with designing a steepener-style coupon rule that monetises curve steepness through the $\alpha(\text{BIR1}_t - \text{BIR2}_t)$ term. The fact that DIR_alt has a lower

average level than DIR anticipates that, under an alternative discounting scenario, cash flows will be discounted at a cheaper funding curve, leading to higher present values and making the note appear even more attractive to investors who can fund close to that curve.

2.3 Interest-rate modelling framework

To simulate the evolution of the interest rates at monthly frequency, the analysis employs one-factor continuous-time short-rate models estimated separately for each curve. For the benchmark rates BIR1 and BIR2, the Vasicek model is adopted:

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t,$$

where κ denotes the speed of mean reversion, θ the long-run mean level, σ the volatility parameter, and W_t a standard Brownian motion. For the discounting curves DIR and DIR_alt, the Cox–Ingersoll–Ross (CIR) model with square-root diffusion is used:

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t} dW_t,$$

which enforces non-negative interest rates and is widely used in interest-rate term-structure modelling.

Maximum-likelihood estimation is performed using the *yuima* package in R, based on the monthly time step $\Delta t = 1/12$. The estimated parameters are summarised in Table 2.

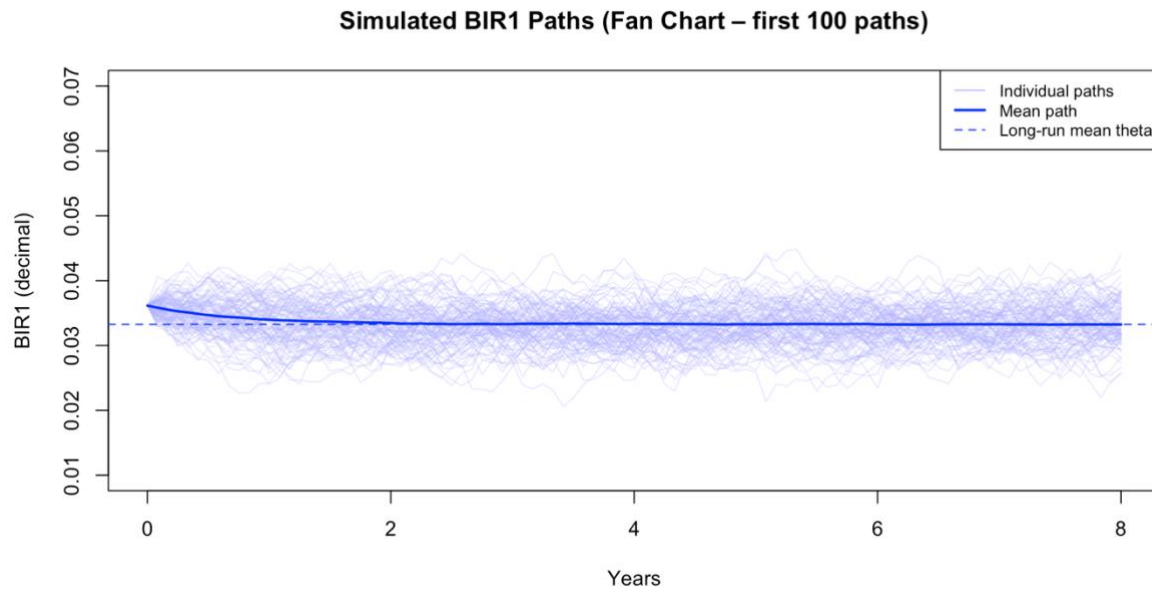
Table 2: Estimated Vasicek/CIR parameters for benchmark and discounting rates.

Series	Model	sigma	kappa	theta
BIR1	Vasicek	0.00508	1.34231	0.03326
BIR2	Vasicek	0.00450	1.37610	0.02221
DIR	CIR	0.02151	0.91046	0.02325
DIR_alt	CIR	0.02353	0.81984	0.01920

For BIR1 and BIR2, the estimated mean-reversion speeds κ are around 1.34–1.38, with long-run means θ close to their historical averages and volatility parameters σ in the 0.45–0.51 per cent range. For DIR and DIR_alt, the CIR estimates imply mean-reversion speeds around 0.82–0.91, long-run means of roughly 2.33 per cent and 1.92 per cent respectively, and higher volatility parameters of about 2.15–2.35 per cent. These parameter estimates imply reasonably fast mean reversion towards historically observed levels combined with volatility that is sufficient to generate economically meaningful dispersion in simulated paths over an 8-year horizon.

To illustrate the dynamics implied by the Vasicek model for BIR1, a fan chart of simulated paths with the mean path and long-run mean can be displayed.

Figure 2: Simulated BIR1 paths (first 100 paths) with mean path and long-run mean.



From a risk-management standpoint, this configuration allows the Monte Carlo engine to capture both the carry from currently observed rates and the uncertainty around future rate realisations that will drive the coupon stream and present-value distribution.

3. Monte Carlo simulation and baseline valuation

3.1 Simulation methodology

The interest-rate dynamics are simulated at monthly frequency over an 8-year horizon, implying $N = 8 \times 12 = 96$ time steps with a step size $dt = 1/12$. For each rate process, 5000 independent Monte Carlo paths are generated, starting from the last observed historical value of the corresponding series, to reflect current market conditions at issuance.

For BIR1 and BIR2, the Vasicek stochastic differential equation is discretised via an Euler–Maruyama scheme:

$$r_{t+dt} = r_t + \kappa(\theta - r_t)dt + \sigma\sqrt{dt} Z_t,$$

where Z_t are independent standard normal shocks. For DIR and DIR_alt, the CIR process is discretised in a similar fashion, with intermediate values floored at zero to preserve non-negativity. Annual coupon dates correspond to every 12 monthly steps; therefore, for the 8-year note there are eight coupon dates at $t = 1, 2, \dots, 8$ years.

For each path and coupon date, the simulated BIR1 and BIR2 values are fed into the coupon formula to compute the capped-and-floored coupon rate c_t , and the corresponding coupon payment is $c_t \times FV$. On the final coupon date, the face value $FV = 100$ is added to the coupon payment. The result is a matrix of cash flows of dimension 8×5000 .

Discount factors are constructed from the simulated DIR paths. For each path, the integral of the short rate is approximated by a discrete sum:

$$\int_0^{t_j} r_s ds \approx \sum_{k=1}^{12j} r_k dt,$$

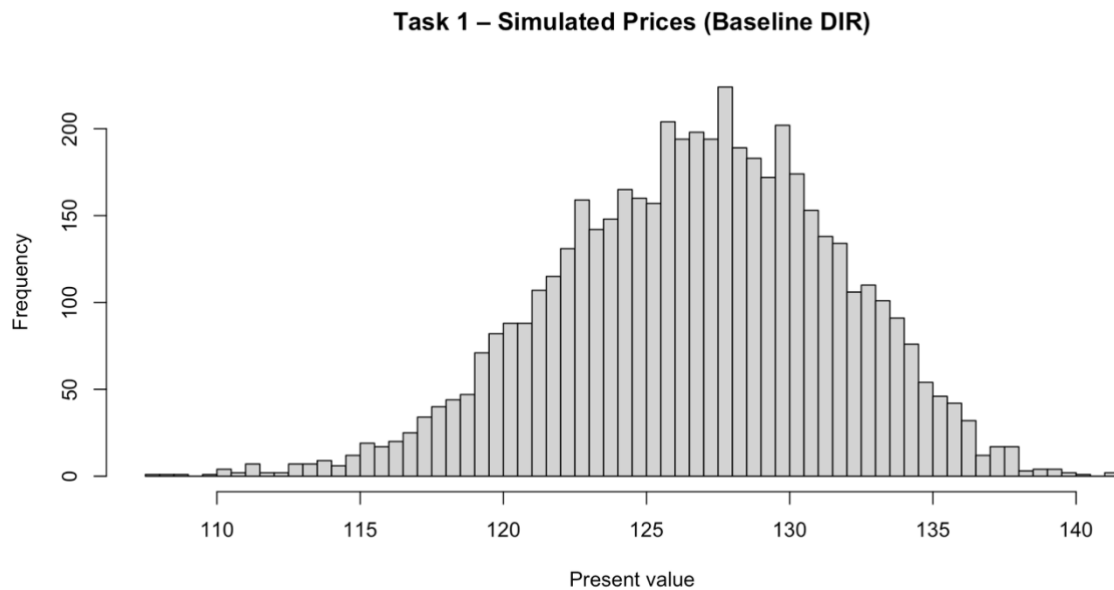
so that the discount factor at coupon date t_j along a given path is $\exp(-\int_0^{t_j} r_s ds)$. Multiplying the cash flows by the appropriate discount factors and summing across coupon dates yields the present value (PV) for each simulation path. The baseline fair value is then estimated as the Monte Carlo mean of these simulated PVs, with a standard deviation, standard error, and confidence interval computed in the usual way. A convergence check is also performed by monitoring how the running mean of the PV stabilises as the number of paths increases from 500 to 5000 in steps of 500. The convergence plot shows that the running mean quickly stabilises around the final estimate, confirming that 5000 paths are sufficient for accurate valuation.

3.2 Baseline valuation results

Under the baseline scenario in which both coupon construction and discounting are based on the simulated BIR1, BIR2 and DIR processes, the estimated Monte Carlo fair value of the 8-year structured note is approximately 126.72. The standard deviation of simulated PVs across the 5000 paths is about 4.93, and the 95 per cent confidence interval for the fair value is roughly [126.59, 126.86]. The note is issued at par (100), so the model-implied fair value indicates a substantial underpricing of about 26.72 units relative to issue price.

The distribution of simulated prices can be visualised by a histogram.

Figure 4: Distribution of simulated present values for the 8-year note under baseline discounting (DIR).



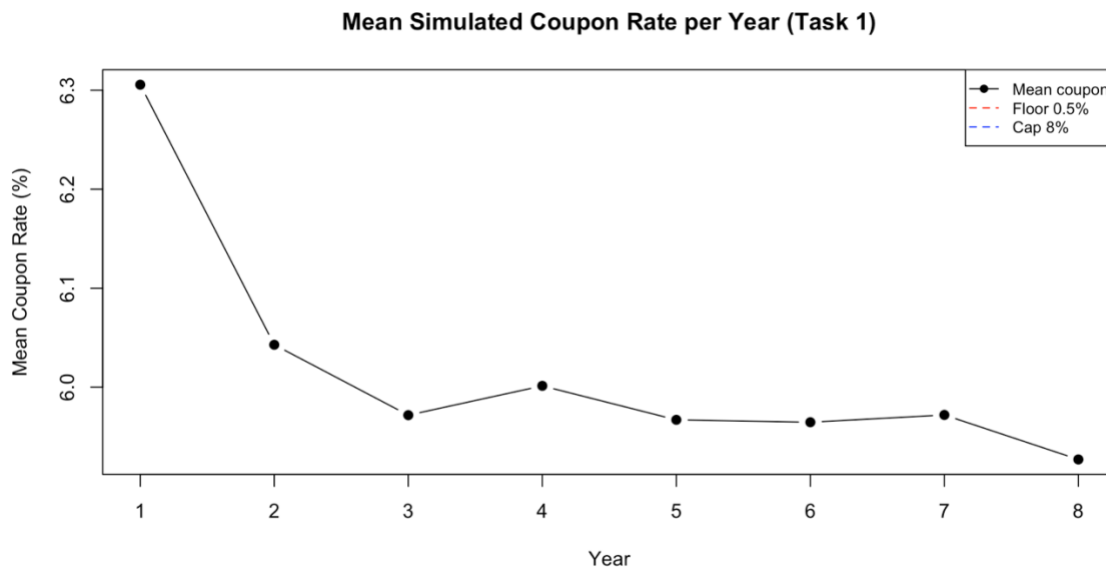
The coupon statistics are also informative. The mean coupon rate across all paths and coupon dates is roughly 6.02 per cent, with a standard deviation of about 1.58 percentage points and a range from the 0.5 per cent floor to the 8 per cent cap. These are summarised in Table 3.

Table 3: Simulated coupon rate statistics for the 8-year note (Task 1, baseline DIR).

Statistic	Value
Mean	0.06019
Std. dev.	0.01578
Min	0.00500
Max	0.08000

To highlight how the expected coupon rate evolves over time, the mean coupon per coupon date can be plotted.

Figure 5: Mean simulated coupon rate per year under the baseline scenario.



The joint evidence from the price distribution and coupon statistics confirms that, under the estimated dynamics of BIR1 and BIR2, the steepener-style formula generates attractive income levels while only occasionally hitting the floor or cap. Investors in the note therefore benefit from a high expected carry profile, in addition to the discount between model value and issue price.

3.3 Baseline investment assessment

From an investment standpoint, the baseline analysis suggests that the 8-year structured note is strongly attractive at par under the calibrated short-rate models. The combination of high expected coupons, principal protection at maturity, and substantial model underpricing creates a favourable risk–return trade-off. The estimated standard deviation of about 4.93 around a mean PV of 126.72 implies that variability in mark-to-model value due to interest-rate movements is modest relative to the expected excess value.

The primary dangers of this situation emerge from its modeling assumptions instead of its Monte Carlo results. Adverse outcomes could materialise if the realised interest rate environment deviates materially from the historical calibration, for example through a structural flattening or inversion of the curve that pushes coupons towards the floor, or through a regime of persistently low DIR that affects discounting differently than assumed. The baseline scenario proves the 8 year note at issuance which shows a clear long recommendation through its modeling framework.

4. Alternative discounting scenario

4.1 Methodological adjustments

Task 2 investigates the sensitivity of the structured note’s valuation to the choice of discount curve by re-pricing the same cash flows using an alternative discounting

interest rate, DIR_alt, instead of DIR. Importantly, the coupon streams are kept identical to those in the baseline scenario: they are still generated from simulated BIR1 and BIR2 using the same Vasicek dynamics and coupon rule parameters. Only the discount factors used to compute present values are altered, now derived from CIR-simulated paths of DIR_alt.

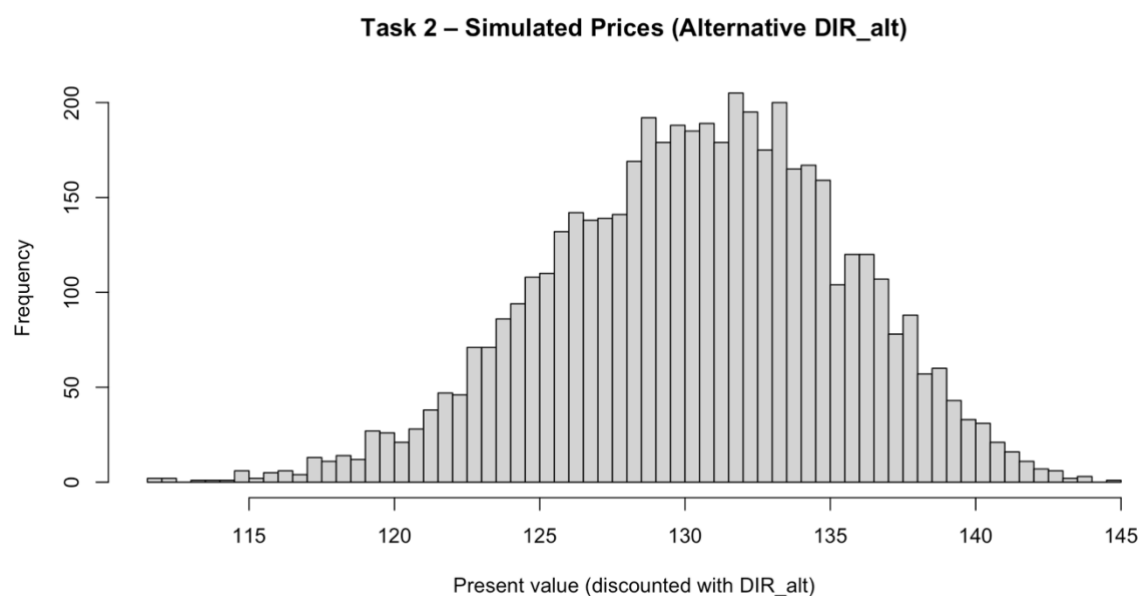
Operationally, the same Euler–Maruyama discretisation is used for the CIR process governing DIR_alt, with parameters estimated from its own historical time series. Discount factors are then computed by exponentiating the negative cumulative sums of DIR_alt along each path, in exact analogy with the baseline case, and applied to the previously constructed cash-flow matrix.

4.2 Valuation impact of lower discount curve

Under the DIR_alt discounting curve, the 8-year note's Monte Carlo fair value rises to approximately 130.39, with a standard deviation of about 5.05 and a very tight 95 per cent confidence interval around this mean. Compared to the baseline fair value of 126.72, the alternative discounting increases the present value by roughly 3.67 units. The underpricing relative to the issue price at par thus expands from about 26.72 to 30.39.

The corresponding distribution of simulated present values can be shown with another histogram.

Figure 6: Distribution of simulated present values for the 8-year note under alternative discounting (DIR_alt).



The economic intuition is straightforward: a lower average discount rate implies a lower required return, so future cash flows are discounted less heavily and therefore acquire a higher present value. Because the structure pays sizeable coupons through time, the sensitivity to discounting is meaningful; a persistent shift in the long-run mean discount rate can produce several points of additional fair value. This illustrates how fixed-income valuations are highly sensitive to funding and collateral assumptions, particularly for instruments with long maturities and coupon streams.

From a hedge-fund perspective, the alternative discounting scenario is particularly relevant for investors who fund themselves close to the DIR_alt curve, for example through OIS-linked repo or other low-cost funding channels. For such investors, the effective hurdle rate for comparing investments is lower, and the note's relative attractiveness becomes even stronger. The scenario therefore reinforces, rather than overturns, the long recommendation from the baseline analysis.

4.3 Role of the discount rate in fixed-income pricing

The contrast between the baseline and alternative discounting scenarios highlights the central role of the discount rate in fixed-income valuation. In both cases, the underlying credit risk and coupon-generation mechanism are identical; only the curve used to discount cash flows changes. A higher discount curve compresses present values and can make even high-coupon instruments appear less attractive relative to par, while a lower curve boosts present values and strengthens the case for investment.

In practice, portfolio managers must carefully align their discounting assumptions with their own funding costs, collateral arrangements, and risk appetites. Using an overly aggressive (too low) discount curve may overstate the attractiveness of risky cash flows, while using a conservative (too high) curve may lead to under-investment in genuinely profitable opportunities. The two scenarios analysed here show that, for this structured note, reasonable variations in discounting still leave a robust excess of fair value over par, suggesting that the investment thesis is not fragile to discount-curve assumptions within the calibrated range.

5. Shorter maturity scenario

5.1 Simulation setup for 4-year note

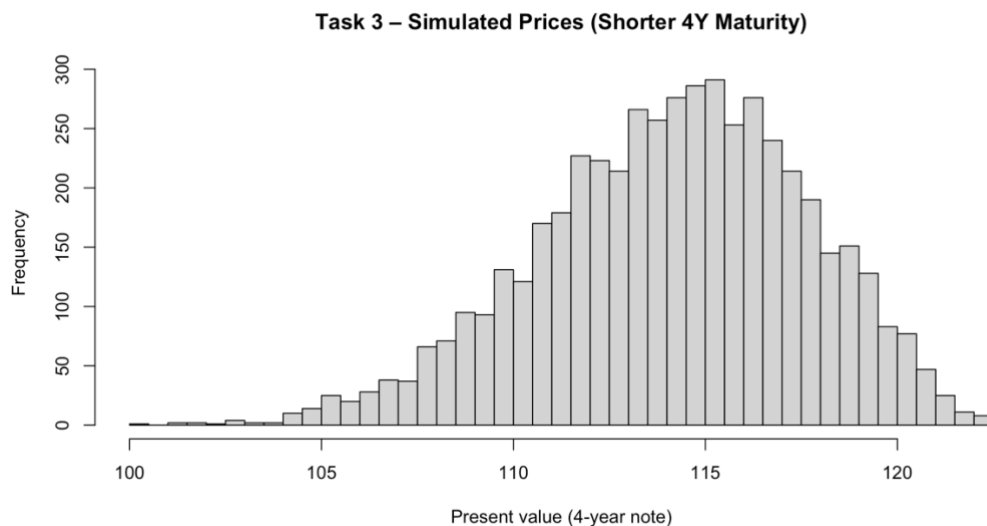
Task 3 explores how shortening the maturity of the structured note affects its valuation and risk profile. The maturity is halved from 8 years to 4 years, implying $N = 4 \times 12 = 48$ monthly steps and four annual coupon dates. The same modelling framework and parameter estimates are retained for BIR1, BIR2, and DIR, ensuring that any change in valuation is driven purely by the shorter cash-flow horizon rather than by recalibration. New Monte Carlo paths are simulated over the 4-year horizon for BIR1, BIR2, and DIR, starting again from the last observed historical values. The coupon rule and parameters ($\alpha = 4$, $\beta = 0.5$, floor 0.5 per cent, cap 8 per cent) remain unchanged, so that the shorter-maturity note is essentially a truncated version of the 8-year structure. The resulting 4-year cash-flow matrix has dimension 4×5000 , and present values are

computed by discounting with simulated DIR in the same manner as in the baseline scenario.

5.2 Valuation and risk for the 4-year note

The Monte Carlo fair value of the 4-year structured note is estimated at approximately 114.13, with a standard deviation of about 3.50 and a 95 per cent confidence interval that is again very tight around the mean. Compared with par, the 4-year note thus appears underpriced by about 14.13 units, which is significantly lower than the 26.72 units of underpricing observed for the 8-year note under baseline discounting. The distribution of simulated present values for the 4-year note can be shown with a separate histogram.

Figure 7: Distribution of simulated present values for the 4-year note under baseline discounting (DIR).



The reduction in underpricing is intuitive: by halving the maturity, the investor forfeits four years of potentially high coupons that contribute substantially to the long-term fair value. At the same time, the standard deviation of the present-value distribution drops from roughly 4.93 to 3.50, reflecting the shorter exposure to interest-rate uncertainty and to the stochastic discount curve. The 4-year note therefore exhibits a more conservative risk–return trade-off: it offers a smaller expected mark-to-model gain but also lower valuation volatility.

5.3 Comparative assessment across scenarios

Table 4: Monte Carlo fair value and risk metrics across valuation scenarios.

Scenario	Maturity (years)	Discount curve	Mean PV	Std. dev.	95% CI low	95% CI high	Excess over par
Task1_8Y_DIR	8	DIR	126.72	4.93	126.59	126.86	26.72
Task2_8Y_DIR_alt	8	DIR_alt	130.39	5.05	130.25	130.53	30.39
Task3_4Y_DIR	4	DIR	114.13	3.50	114.03	114.22	14.13

All three scenarios show the note trading below its model-implied fair value at issuance, but the magnitude of underpricing and the associated risk levels vary meaningfully. The alternative discounting scenario strengthens the long case for the 8-year note, while the shorter-maturity scenario offers a scaled-down but still positive opportunity for more risk-averse mandates. These patterns emphasise that maturity and discount-curve choices are key levers in structuring trades that align with specific portfolio objectives.

5.4 Risk decomposition and portfolio perspective

From a portfolio-management perspective, the structured note's risk profile can be decomposed into three main dimensions.

First, *level risk* arises from movements in the overall discount curve (DIR or DIR_alt), which directly affect the present value of all future cash flows through the stochastic discount factors. Second, *slope risk* is embedded in the coupon rule via the $\alpha(\text{BIR1}_t - \text{BIR2}_t)$ term, creating exposure to changes in the spread between the two benchmark rates and thereby to curve-steepening or curve-flattening scenarios. Third, *horizon risk* reflects the choice of maturity, with the 8-year structure carrying more duration and convexity than the 4-year variant, and thus greater sensitivity to shocks in both level and slope.

A hedge fund can maintain its required slope exposure through interest-rate swaps or futures, which it uses to hedge their level risk while managing its net duration according to its established risk limits. The 8-year note serves as an appropriate investment for investors who accept extended time periods of interest rate and curve risk, whereas the 4-year note functions as a shorter-duration investment or as an essential component for advanced relative-value trading methods.

6. Conclusion: implications for the investment decision

The Monte Carlo analysis of the interest-rate-linked structured note leads to several clear conclusions from a hedge-fund perspective. First, under the baseline calibration of BIR1, BIR2, and DIR, the 8-year note appears materially underpriced at par, with an estimated fair value around 126.72 and a relatively modest valuation volatility. The steepener-style coupon rule generates attractive expected coupons centred around 6 per cent, and principal repayment at maturity further enhances the defensive characteristics of the structure.

Second, when the same cash flows are discounted with an alternative, lower discount curve (DIR_alt), the fair value increases to roughly 130.39, reinforcing the conclusion that the note is attractive for investors whose funding and collateral terms more closely resemble this cheaper curve. This scenario illustrates the strong role that discounting assumptions play in fixed-income valuation and explains why different investors may have different assessments of the same instrument depending on their cost of capital. Third, shortening the maturity to 4 years reduces both the note's underpricing and its valuation volatility, effectively offering a lower-duration version of the trade. The 4-year note still appears underpriced at par, but by a smaller margin of about 14.13 units, making it suitable for mandates with tighter risk limits or shorter investment horizons that nonetheless seek positive carry and mark-to-model upside.

The evidence in all scenarios demonstrates that investors follow a sustained bias towards the structured note that existed at its issuance date. Investors who can handle higher interest rate and curve risk should select the 8 year version of the structured note while more risk averse investors should consider the 4 year version. A hedge fund can execute the trade by using its risk budget to determine the 8 year position size and then using either the short-term maturity assets or hedging tools to adjust its duration and curve risk.

Limitations and model risk

The report findings depend on the particular modelling methods and calibration period which were selected for the study. The interest-rate dynamics are represented by single-factor Vasicek and CIR models estimated on historical monthly data without allowing for jumps regime shifts or time-varying volatility; in reality monetary policy regimes and macroeconomic shocks create patterns of behavior which these models fail to fully capture. The analysis excludes credit and liquidity risk while it treats funding costs as operational at the selected discount rates while it overlooks expenses related to transactions. As a result, the reported fair values should be interpreted as model-based benchmarks rather than exact market predictions, and prudent portfolio management would complement them with stress tests and scenario analysis before implementing large positions.